Quantum State Discrimination with Bosonic Channels and Gaussian States

Course: PH3203 Advanced Quantum Mechanics
Instructor: Professor Sourin Das

A Term Paper Review of Chapters 1–3[1]

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Abstract

[ABSTRACT]

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3.1 Distinguisability Of States

Although, the problem of distinguisability is defined for any number of states, let us start with the problem of distinguisability of two states, $\hat{\rho}_0$ and $\hat{\rho}_1$, with their corresponding equiprobable binary hypothesis H_0 and H_1 respectively. However, so far, we have only discussed more of the formalism, but nothing that could be used quantify the distinction between two states. In particular, we want something that can quantify some *distance* between the two states or some kind of method to quantify some *error* for distinction between two states given the measurement. For the notion of *distance*, we use *fidelity of states* and as discused in chapter 1, the *Helstrom's Bound* can be viewed as a measure of the error which gives the minimum probability of error and the corresponding POVM that gives the minimum measurement errors.

3.1.1 Helstrom's Optimal Discrimination

As discussed in chapter 1, Helstrom's Bounds corresponds to the generic optimization problem to find the POVM that gives the best measurement with minimum error probability. Given the two states, we consider a dichotomous POVM $\{\hat{E}_0, \hat{E}_1\}$, where $\hat{E}_0$ corresponds to the hypothesis $\hat{H}_0$ and $\hat{E}_1$ corresponds to the hypothesis $\hat{E}_1$ respectively. The error probability can simply be obtained from the hypothesis testing for H_0 and H_1 as

$$P_E = \frac{1}{2}(P(H_0|H_1) + P(H_1|H_0)) \tag{1}$$

where,
$P(H_0|H_1)$ is the probability of concluding that H_0 is true, when H_1 is and
$P(H_1|H_0)$ is the probability of concluding that H_1 is true, when H_0 is true and
the factor of $\frac{1}{2}$ arises from the equiprobable assumptions.

Also, note that the expectation of an operator $\hat{O}$ given the system is in state $\hat{\rho}$ is given by $\left\langle \hat{O} \right\rangle = tr(r\hat{h}o\hat{O})$. So, if the system is in $\hat{\rho}_j$ and we make a measurement with the operator with $\hat{E}_i$, then the probability of error in the measurement is given by

$$P(H_i|H_j) = tr(\hat{\rho}_j \hat{E}_i) \tag{2}$$

Thus, using eq 2 in eq 1, we get

$$P_E = \frac{1}{2}(tr(\hat{\rho}_1\hat{E}_0) + tr(\hat{\rho}_0\hat{E}_1)) \tag{3}$$

Also, since we defined our POVM to be the set $\hat{E}_0, \hat{E}_1$ and POVM are complete, so $\hat{E}_0 + \hat{E}_1 = I \implies \hat{E}_0 = I - \hat{E}_1$. So, eq 3 will be,

$$\begin{aligned} P_E &= \frac{1}{2}(tr(\hat{\rho}_1(I - \hat{E}_1)) + tr(\hat{\rho}_0\hat{E}_1)) \\ &= \frac{1}{2}(tr(\hat{\rho}_1) - tr(\hat{\rho}_1\hat{E}_1) + tr(\hat{\rho}_0\hat{E}_1)) \\ &= \frac{1}{2}(1 - tr(\hat{\rho}_1\hat{E}_1) + tr(\hat{\rho}_0\hat{E}_1)), \text{since } tr(\hat{\rho}_i) = 1 \\ &= \frac{1}{2}(1 - tr((\hat{\rho}_1 - \hat{\rho}_0)\hat{E}_1)) \end{aligned} \tag{4}$$

Note, all we aim to is find the correct POVM set so that the error probability is minimum. For this case, as $\hat{E}_0 + \hat{E}_1 = I$, so, we can express $\hat{E}_0$ in terms of $\hat{E}_1$. Thus, we have to minimize P_E over the measurement operator $\hat{E}_1$. Define $\hat{\gamma} = \hat{\rho}_1 - \hat{\rho}_0$. Clearly, $tr(\hat{\gamma}) = tr(\hat{\rho}_1) - tr(\hat{\rho}_0) = 1 - 1 = 0$. Since, both $\hat{\rho}_0$ and $\hat{\rho}_1$ are both positive and semi definite, so the eigenvalues of $\hat{\gamma}$ can be both positive or negative. Since both $\hat{\rho}_0$ and $\hat{\rho}_1$ are hermitian, so $\hat{\gamma}$ is also hermitian, hence it has the spectral decomposition,

$$\hat{\gamma} = \sum_k \mu_k |k\rangle \langle k| \tag{5}$$

where μ_k are the real eigenvalues. Now expanding $\hat{E}_1$ in the same basis

$$\hat{E}_1 = \sum_{k,l} e_{kl} |k\rangle \langle l| \tag{6}$$

Hence,

$$\begin{aligned} tr(\hat{E}_1\hat{\gamma}) &= tr\left(\sum_{k,l} e_{kl} |k\rangle \langle l| \cdot \sum_m \mu_m |m\rangle \langle m|\right) \\ &= \sum_{klm} e_{kl}\mu_m \langle l|m\rangle \langle m|k\rangle \\ &= \sum_{klm} e_{kl}\mu_m \delta_{l,m}\delta_{m,k} \\ &= \sum_k e_{kk}\mu_k \end{aligned} \tag{7}$$

Since POVMs are positive valued, so e_{kk} is positive. Now, μ_k can be positive or negative, but since we are optimizing e_{kk}, if μ_k is negative, it will lead to negative propagation

of the errors, thus for such cases e_{kk} must be 0. Hence the net contribution will come only from the positive e_{kk} values. Thus the optimal $\hat{E}_1$ is one that projects onto positive part of $\hat{\gamma}$, $\hat{\gamma}_+$ (say) which projects to the subspace by the eigenstates of $\hat{\gamma}$ with positive eigenvalues. Thus, we have $tr(\hat{\gamma}\hat{E}_1) = tr(\hat{\gamma}_+) = \frac{1}{2}||\hat{\gamma}||_1$ where $||\hat{\gamma}||_1 = tr(|\hat{\gamma}|) = tr(\sqrt{\gamma^\dagger\gamma})$. Thus, using this, eq 4 can be written as

$$P_E^{min} = \frac{1}{2}(1 - ||\hat{\rho}_1 - \hat{\rho}_0||_1) \tag{8}$$

However, the calculation of the minimum probability of error using the above hypothesis is only for a single particle system with two states. But, the same analysis and calculations for an N particle system would be very difficult. This is because as the number of particles in the system increases the Hilbert Space grows exponentially over N as the N copy density matrix resides in the tensor product state of dimension d^N, where d is the dimension (degrees of freedom) for a single particle state, which may not simplify for larger N values. Moreover, opimization of such d^N matrix set (POVM) would also be analytically as well as numerically very difficult. However, one can calculate the *Quantum Chernoff Bounds*, which gives the asymptotic dependency of N copys of the system on the probability of error. Then the quantum hypothesis test becomes

$$\begin{aligned} H_0 &: \hat{\rho}_0 \otimes ... \otimes \hat{\rho}_0 \\ H_1 &: \hat{\rho}_1 \otimes ... \otimes \hat{\rho}_1 \end{aligned} \tag{9}$$

Thus for this case, the minimum probability of error is given by

$$P_{E,N}^{min} = \frac{1}{2}(1 - ||\hat{\gamma}_N||_1) \tag{10}$$

where $\hat{\gamma}_N = \hat{\rho}_1 \otimes ... \otimes \hat{\rho}_1 - \hat{\rho}_0 \otimes ... \otimes \hat{\rho}_0$

3.1.2 Fidelity

Fidelity between two states could be used as a notion of measure of the *distance* between two states. For two states, $\hat{\rho}_1$ and $\hat{\rho}_2$, fidelity is defined as,

$$F(\hat{\rho}_1, \hat{\rho}_2) = \left(tr\left(\sqrt{\sqrt{\hat{\rho}_1}\hat{\rho}_2\sqrt{\hat{\rho}_1}} \right) \right)^2 \tag{11}$$

- F($\hat{\rho}_1$, $\hat{\rho}_2$) ≤ 1
- F($\hat{\rho}_1$, $\hat{\rho}_2$) = F($\hat{\rho}_2$, $\hat{\rho}_1$)
- If $\hat{\rho}_1$ is a pure state, $\hat{\rho}_1 = |\psi_1\rangle\langle\psi_1|$, then F($\hat{\rho}_1$, $\hat{\rho}_2$) = $\langle\psi_1|\hat{\rho}_2|\psi_1\rangle$
- F($\hat{\rho}_1$, $\hat{\rho}_2$) is invariant under unitary transformations on the state space.

- $\mathrm{F}(\hat{\rho}_1 \otimes \hat{\sigma}_1,\, \hat{\rho}_2 \otimes \hat{\sigma}_2) = \mathrm{F}(\hat{\rho}_1,\, \hat{\rho}_2)\ \mathrm{F}(\hat{\sigma}_1,\, \hat{\sigma}_2)$

3.1.3 Quantum Chernoff Bounds

So far, we have discussed how the error is quantified and methods of quantifying the error. But, how is the actual minimization of the errror probability occurs. Chernoff solved this problem in the asymptotic regime and gave the asymptotic dependence on the probability of error if we have N copies of the system, i.e. a N trail hypothesis testing.[2] He showed that the probability of error decreases exponentially with the number of tests performed N, i.e. $P_E \approx exp(-\kappa N)$. The optimal exponent κ arising in the asymptotic limit is called Chernoff Bounds[2]. But why is Chernoff Bounds even considered... This may go back to 'One of the virtues of the Chernoff bound is that it yields a very natural distance measure between probability distributions; it is essentially the unique distance measure in the ubiquitous situation of independent and identically distributed (i.i.d.) random variables.'[2]

The quantum chernoff bounds serves as the upper limit the the minimum probability of error 10, i.e.

$$P_{N,E}^{min} \leq P_{QC}^{N} = \frac{1}{2} exp(-\kappa N) \tag{12}$$

where $\kappa = -ln\left(\min_{\,0\leq s\leq 1} tr(\hat{\rho}_0^s \hat{\rho}_1^{(1-s)})\right)$

Before diving deep into the quantum chernoff bounds, let us discuss a bit of the classical chernoff bounds. Consider a random variable X and $a \in \mathbb{R}$. Then, [3]

$$\begin{aligned} P(X \geq a) &\leq min_{s>0} e^{-sa} M_X(s) \\ P(X \leq a) &\leq min_{s<0} e^{-sa} M_X(s) \end{aligned} \tag{13}$$

For the optimal decision, we define the log likelihood,

$$L(x) = log \frac{p_1(x)}{p_0(x)}$$

For N trial observations

$$X = \sum_{i=1}^{N} L(x_i) = \sum_{i=1}^{N} log \frac{p_1(x_i)}{p_0(x_i)}$$

where we declare H_1 is true if X ¿ 0 else we declare H_0 is true. Now we refer to the eq 1, which using above under hypothesis H_1

$$P(H_0|H_1) = P(X \leq 0|H_1) \leq e^{-s\cdot 0} M_X(s) = M_X(s)$$

where $s < 0$ The moment generating function $M_X(s)$ is defined as $\langle e^{sX} \rangle$, Hence

$$\begin{aligned} M_X(s) = \langle e^{sX} \rangle &= \langle e^{s \sum_{i=1}^{N} log \frac{p_1(x_i)}{p_0(x_i)}} \rangle \\ M_X(s) &= \langle e^{\sum_{i=1}^{N} log \frac{p_1(x_i)}{p_0(x_i)}^s} \rangle \\ M_X(s) &= \langle \prod_{i=1}^{N} e^{log \frac{p_1(x_i)}{p_0(x_i)}^s} \rangle \\ M_X(s) &= \prod_{i=1}^{N} \langle \frac{p_1(x_i)}{p_0(x_i)}^s \rangle \end{aligned} \tag{14}$$

Since, each factor is identical and $\langle f \rangle = \sum_{x_i} p(x) f(x)$ so,

$$M_X(s) = \left(\sum_x p_1(x) \left(\frac{p_1(x)}{p_0(x)} \right)^s \right)^N = \left(\sum_x p_1(x)^{1+s} p_0(x)^{-s} \right)^N \tag{15}$$

Taking $s \to -s$, i.e. $s > 0$

$$M_X(s) = \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{16}$$

$$P(H_0|H_1) \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{17}$$

Similarly for $P(H_1|H_0)$, we get

$$P(H_1|H_0) \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{18}$$

where $s > 0$ Combining eq 17 and eq 18 with eq 1

$$P_E^N \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{19}$$

Notice this takes the same form as the quantum Chernoff Bounds upto a factor of $\frac{1}{2}$, where instead of the probability desity functions $p_0(x)$ and $p_1(x)$, we have the density matrices ρ_0 and ρ_1.

One of the notable observation is that the Chernoff Bounds are exponentially tight. As the number of trails increases i.e. $N \to \infty$, $ln(P_{N,E}^{min})/N \to -\kappa$

Also, defining $Q_s = tr(\hat{\rho}_0^s \hat{\rho}_1^{1-s})$ and $Q = \min_{0 \leq s \leq 1} Q_s$, the quantum Chernoff Bounds can be written as

$$P_{QC}^N = \frac{1}{2} Q^N \tag{20}$$

The aim to find the minimum error would require the minimization of Q_s over all valures of $s \in [0,1]$. Setting s = $\frac{1}{2}$, we will arrive at the *Quantum Bhattacharyya Bound*, however this bound is not assymptotically tight as the Quantum Chernoff Bound.

$$P_B^N = \frac{1}{2}\left(tr(\sqrt{(}\hat{\rho}_0\hat{\rho}_1))\right)^N \tag{21}$$

If the two states are indistinguisable, i.e. $\hat{\rho}_0 = \hat{\rho}_1$, then $P_{QC}^N \approx P_B^N$ and

$$\frac{1-\sqrt{1-F(\hat{\rho}_0,\hat{\rho}_1)}}{2} \leq P^{(1)} \leq P_{QC}^{(1)} \leq \frac{\sqrt{F(\hat{\rho}_0,\hat{\rho}_1)}}{2}$$

Also, it is noted that Q_s is a convex function over s and joint-concave over $\hat{\rho}_0$ and $]r\hat{h}o_1$. Q_s can be written as

$$Q_s = tr(\hat{\rho}_0^s]r\hat{h}o_1^{1-s}) = \sum_k e_k^{(0)^s} e_k^{(1)^{1-s}}$$

where $e_k^{(0)}$ and $e_k^{(1)} \geq 0$ Thus each element of Q_s is $f_k(s) = a_k^s b_k^{1-s} = e^{slna_k} e^{(1-s)lnb_k}$, taking the 2nd derivative of f_k w.r.t. s, we see $f_k^{''}(s) = (lna_k - lnb_k)^s \cdot f_k(s) \geq 0$. Hence Q_s is convex over s. Also using Lieb's Theorem [4]

Theorem 3.1. *Lieb's Theorem: Let K be a fixed matrix in $\mathbb{C}^{n\times m}$. Then for any $t \in [0,1]$, the map*

$$(A,B) \to tr[K^* A^{1-t} K B^t]$$

is jointly concave in (A, B) where A and B are respectively $n \times n$ and $m \times m$ Hermitian positive definite matrices.

Taking $A = \hat{\rho}_0$ and $B = \hat{\rho}_1$, and $K = \mathbb{I}$ and $t = s$, then Q_s is concave over $\hat{\rho}_0$ and $\hat{\rho}_1$.

3.2 Fidelity for Gaussian States

3.2.1 Single Mode Gaussian States

For two single mode gaussian thermal states with mean 0 and covariance matrices σ_1 and σ_2 corresponding to states $\hat{\rho}_1$ and $\hat{\rho}_2$ is given by

$$F_0(\hat{\rho}_1,\hat{\rho}_2) = \frac{2}{\sqrt{det(\sigma_1+\sigma_2)}(\sqrt{det(\theta)} - \sqrt{det(\theta)-1})} \tag{22}$$

where,
$det(\theta) = 1 + \frac{P}{det(\sigma_1+\sigma_2)}$
$P = (det(\sigma_1)-1)(det(\sigma_2)-1)$
It was also derived in the paper, the fidelity for a two single mode displaced squeezed thermal states. By the Williamson's Theorem, it could be seen that any arbitrary

single mode gaussian state can be expressed as a displaced squeezed thermal state, i.e. $\hat{\rho}_G = \hat{U}_S \hat{\rho}_{th} \hat{U}_S^\dagger$. Thus, the fidelity for any single mode gaussian state,

$$F(\hat{\rho}_1, \hat{\rho}_2) = F_0(\hat{\rho}_1, \hat{\rho}_2) exp\left(-\frac{1}{2}(\overline{R_1} - \overline{R_2})^T \kappa (\overline{R_1} - \overline{R_2})\right) \tag{23}$$

where $F_0(\hat{\rho}_1, \hat{\rho}_2)$ is defined as eq 22 and $\kappa = \mathbb{I} - (\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)$

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